

CONTINUAL LEARNING FRAMEWORK FOR ADAPTIVE IOT ROBOTS IN DYNAMIC ENVIRONMENTS

GROUP 12

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DEGREES OF FREEDOM(DOF)

- Degrees of freedom in robotics are the number of independent movements a robot can make, either by shifting position or rotating around an axis. Robots move in two main ways: translational (straight line) and rotational (spinning or turning).
- Each joint or axis adds one degree of freedom, similar to how your shoulder, elbow, and wrist work together to move your arm. A 6-axis robot arm, for instance, can bend, reach, and rotate to handle complex tasks. From industrial arms to drones, understanding DOF explains why some robots move with surprising precision while others feel more limited.
- **7 DOF (Seven degrees of freedom):** The seventh degree of freedom offers more flexibility by allowing alternative paths to reach the same point. It's similar to how your arm can adjust slightly to reach around an object or avoid a collision.

INVERSE KINEMATICS

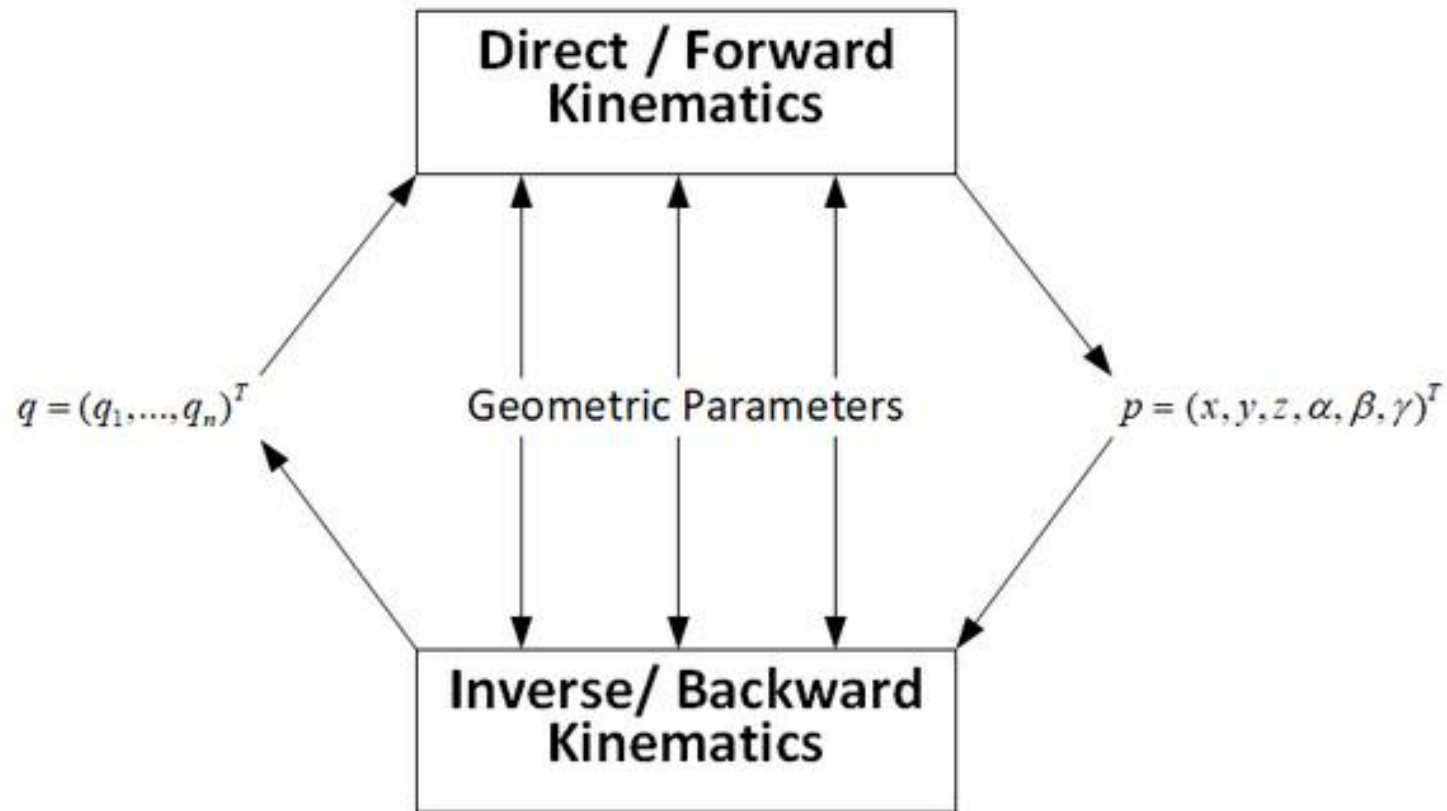
Inverse Kinematics is the calculation of the variables of the set of joints and linkages connected to an end effector. Given the position and orientation of the end effector, inverse kinematics can be used to calculate the variables regarding those joints and linkages including position, angle, and orientation.

The appropriate path to move from one position to another can be calculated using a Jacobian matrix.

Joint Angle

Transformation

Cartesian Coordinates



FEEDFORWARD NEURAL NETWORK(FNN) / MULTILAYER PERCEPTRON(MLP):

- **Definition:** A foundational artificial neural network where data moves in **one direction**—from input, through hidden layers, to output—without any feedback loops.
- **Structure:** Composed of at least three layers (Input, Hidden, Output). Neurons are fully connected to the next layer.
- **Key Components:**
 - **Input Layer:** Receives raw features.
 - **Hidden Layer(s):** Performs computations (weighted sum + activation function) to learn non-linear patterns.
 - **Output Layer:** Generates the final prediction.
- **Learning Process (Training):** Uses **Backpropagation** and Gradient Descent to minimize error by adjusting weights and biases.
- **Common Activation Functions:** Sigmoid, Tanh, ReLU.
- **Applications:** Classification (e.g., Image recognition), Regression analysis.

PROBLEM STATEMENT

- Manual pick and place is inefficient.
- Industries need automation with high precision.
- Our aim is to develop a remotely operated robotic pick-and-place system.

OBJECTIVES

- Design 7 DOF revolute robot.
- Implement DH modeling.
- Compute IK at each step of movement of arm.
- Enable teleoperation using mqtt.

COURSE MAPPING

- Unit1 Teleoperation
- Unit2 Reactive Paradigm
- Unit3 Transformation matrices
- Unit4 DH(Denavit-Hartenberg), FK(Forward Kinematics) and IK(Inverse Kinematics)

SYSTEM ARCHITECTURE

- User enters the colour and place coordinates on new terminal using mqtt broker - mosquito.
- Data is sent to MQTT Server.
- Backend parses inputs and triggers IK computation.
- Computed joint angles are sent to the robot.
- Animation/movement of robot motion.
- User observes real-time movement on simulator window.

ROBOT MODEL

Joint Index	Joint Type	Description
0	Revolute	Base rotation
1	Revolute	Shoulder pitch
2	Revolute	Shoulder roll
3	Revolute	Elbow rotation
4	Revolute	Wrist pitch
5	Revolute	Wrist roll
6	Revolute	Wrist yaw / end-effector orientation

DH PARAMETERS

- Denavit-Hartenberg convention is used for modeling.
- Each joint is represented by four parameters θ (joint angle), d (link offset), a (link length), α (link twist/angle).
- DH table is created for all three links.
- Homogeneous transformation matrices are derived.
- These matrices represent rotation and translation between frames.

EXPECTED RESULTS

- Robot successfully moved object from source to target.
- End-effector followed smooth and collision-free path.
- Positioning error was minimal.
- System response time was within acceptable limits.

FUTURE SCOPE

- Integration with physical robotic arm.
- Addition of vision sensors for object detection.
- Obstacle avoidance using path planning algorithms.